part2pop: a modular framework for standardizing and analyzing aerosol particle populations

Summary

Aerosol particles are chemically diverse, multicomponent systems that vary widely in size, composition, and mixing state. These particle-scale differences strongly influence cloud condensation nuclei activity, ice nucleation, and optical properties (Riemer et al. 2019; Petters and Kreidenweis 2007; Cubison et al. 2008; Fierce et al. 2013, 2016, 2020; Alpert and Knopf 2016; Lata et al. 2021).

Across models, observations, and parameterized descriptions, aerosol populations are represented in incompatible ways. Different representations preserve different aspects of aerosol size, composition, and mixing state, which makes aerosol populations difficult to compare directly (Riemer et al. 2009; Liu et al. 2016; Tilmes et al. 2023; Stevens and Dastoor 2019; Fierce et al. 2024). As a result, similar analyses are frequently reimplemented for each representation, making comparison and reuse time-consuming.

`part2pop` is a lightweight Python library is a lightweight Python library that standardizes heterogeneous aerosol populations into a common object model. The package supports population construction from model output, parameterized distributions, and observation-constrained reconstructions, and it provides reusable particle- and population-level diagnostics for aerosol composition, hygroscopicity, cloud activation, optical properties, and freezing-oriented analyses. Its builder/registry design makes the package extensible, allowing new species definitions, population types, particle morphologies, and diagnostic variables to be added without modifying the core representation.

Statement of need

Aerosol populations from particle-resolved models, reduced representations such as modal and sectional models, parameterized distributions, and observation-based reconstructions represent different information about particle size, composition, abundance, and mixing state (Riemer et al. 2009; Zaveri et al. 2008; Bauer et al. 2008; Liu et al. 2016; Tilmes et al. 2023; Mann et al. 2012; Stevens and Dastoor 2019; Jayne et al. 2000; DeCarlo et al. 2006; Wang, Pikridas, Spielman, et al. 2017; Zelenyuk et al. 2015). Consequently, these populations are not directly comparable, and analyses such as composition metrics, CCN activity, optical properties, and freezing diagnostics are often reimplemented separately for each representation.

`part2pop` addresses this gap by providing a common aerosol-population interface that makes such analyses reusable across heterogeneous sources. The package is not a new aerosol simulation model. Instead, it standardizes populations derived from simulations, observations, and parameterized distributions so that they can be analyzed, compared, and reused consistently. Intended users include aerosol model developers, instrument-developers and observationalists, process modelers requiring aerosol inputs, instructors and students, and researchers performing intercomparisons or sensitivity analyses. The need addressed is not a lack of aerosol-physics models, but a lack of a shared aerosol population representation that makes existing tools reusable across models, parameterizations, and observations.

State of the field

Existing software in aerosol science typically focuses on either source-specific aerosol populations or targeted downstream calculations, rather than interoperability across representations. Aerosol simulation frameworks generate populations in source-specific representations, reflecting trade-offs between physical detail and computational efficiency (Riemer et al. 2009; Liu et al. 2016; Tilmes et al. 2023; Mann et al. 2012). Observational workflows constrain different subsets of aerosol population properties through instrument-specific measurements (Wang, Pikridas, Spielman, et al. 2017; Zelenyuk et al. 2015; DeCarlo et al. 2006). Other tools then target specific diagnostics such as optical properties, cloud activation, or ice nucleation (Sumlin et al. 2018; Petters and Kreidenweis 2007; Alpert and Knopf 2016). However, these components do not share a common representation of aerosol populations, limiting reuse across sources. As a result, populations derived from different sources cannot be readily combined, compared, or analyzed using shared diagnostics.

`part2pop` complements rather than duplicates these tools. Its distinct contribution is a shared representation layer that standardizes heterogeneous aerosol populations and enables the same downstream diagnostics to be applied across them.

Software design

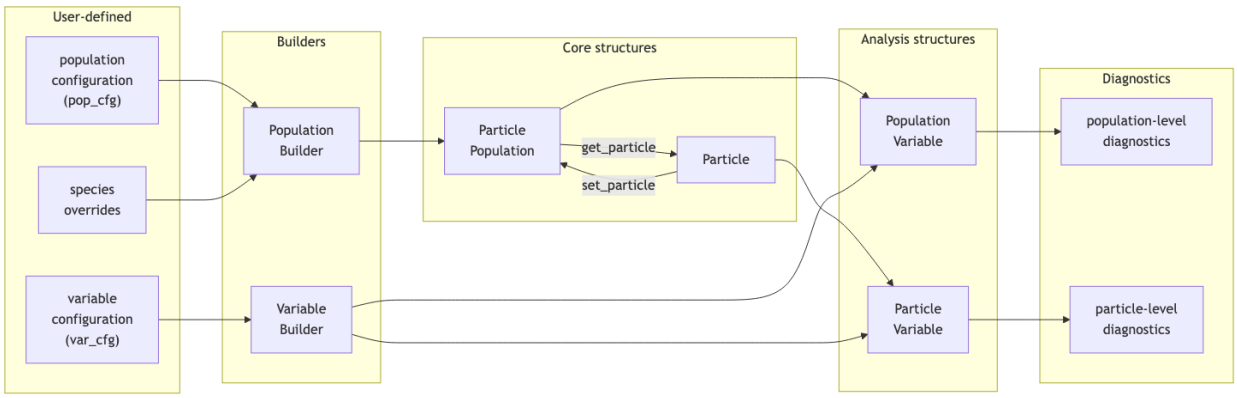

Figure 1: Overview of the config-driven architecture in `part2pop`. User-defined configuration files are processed by builder modules to construct standardized particle populations and derived analysis variables. Particle- and population-level diagnostics are then computed from these shared representations.

`part2pop` adopts a config-driven architecture centered on a small set of core data structures and registry-backed builders (Figure 1). The primary structures are `AerosolSpecies`, `Particle`, and `ParticlePopulation`. `AerosolSpecies` defines intrinsic species properties such as density, molecular weight, hygroscopicity, and refractive-index-related defaults, with optional overrides applied at build time. `ParticlePopulation` stores a species list, species-resolved particle masses, number concentrations, and particle identifiers. Individual particles are accessed and modified through the population interface. Together, these structures provide a standardized representation of aerosol composition, size, and abundance across heterogeneous sources.

The core representation is mass-based. This is a deliberate design trade-off: it is general enough to accommodate heterogeneous population sources while remaining simple enough to support consistent derived-property calculations. Population construction is handled by `PopulationBuilder` using `population_cfg`, and diagnostic construction is handled by `VariableBuilder` using `var_cfg`. Both rely on registry-backed factories, so new population builders and diagnostic modules can be added without modifying the core classes.

Current population types include model-derived populations such as `partmc` and `mam4`, parameterized populations such as `monodisperse`, `binned_lognormals`, and `sampled_lognormals`, and observation-based populations such as `hiscale_observations`. The `hiscale_observations` builder reconstructs populations

from complementary measurements associated with the HI-SCALE field campaign, including FIMS size distributions, AIMMS aircraft-state data, miniSPLAT composition classes, and AMS bulk-composition constraints (Fast et al. 2019; Wang, Pikridas, Spielman, et al. 2017; Wang, Pikridas, Pinterich, et al. 2017; Beswick et al. 2008; Zelenyuk et al. 2015; Jayne et al. 2000; DeCarlo et al. 2006).

Process-specific extensions such as `OpticalPopulation` and `FreezingPopulation` wrap the same base population representation rather than duplicating population logic. This allows optical and freezing calculations to reuse the same underlying particle and population definitions while interfacing with established frameworks such as κ-Köhler theory and existing optics software (Petters and Kreidenweis 2007; Sumlin et al. 2018; Alpert and Knopf 2016). A visualization module and optional graphical viewer support exploratory analysis.

The architecture is also intentionally designed to be modifiable by both human developers and automated code-generation systems. Config-driven workflows, registry-based module discovery, minimal coupling between core data structures and downstream diagnostics, and stable builder interfaces localize code changes. As a result, new population builders or diagnostics can be added without invasive edits to the core package, lowering the barrier to extension for both human developers and AI-assisted code generation.

Research impact statement

`part2pop` targets an active methodological bottleneck in aerosol science: representation choice itself can introduce structural error in aerosol diagnostics and can complicate comparison across models and datasets (Fierce et al. 2017, 2024; Fierce and McGraw 2017). By making particle-resolved, modal, sectional, parameterized, and observation-based populations analyzable through a shared interface, the package provides reusable infrastructure for cross-representation comparison, diagnostic reuse, and downstream model initialization.

The current release includes reproducible workflows for optical, CCN-relevant, freezing-oriented diagnostics for particle populations from a variety of sources. This positions `part2pop` as reusable research infrastructure rather than a study-specific analysis script and supports its use in intercomparison and observation-constrained aerosol analysis.

AI usage disclosure

The core architecture, data model, and software design were developed by the authors without the use of generative AI tools. After the core structure was established, generative AI tools (ChatGPT 5.1–5.4) were used for code development and refactoring, development of the graphical user interface, and drafting and editing of documentation and manuscript text.

All AI-generated code and text were reviewed, modified, and verified by the authors, validated through testing and example workflows, and checked for scientific correctness. The software architecture was intentionally designed to support modification by automated code-generation tools, including large language models, through its modular and configuration-driven structure.

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